

$$V(r) = \infty \quad r < r_0$$

$$= -\frac{\hbar^2}{\mu r^2} \quad r > r_0$$

2) $R_{k1}(r)$ must satisfy $l=1$ Schrödinger equation

$$\left(-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + V(r) + \frac{\hbar^2 l(l+1)}{2\mu r^2} \right) u_{k1}(r) = \overset{\hbar^2 k^2/2\mu}{E} u_{k1}(r)$$

$$R_{k1}(r) \cdot r = u_{k1}(r)$$

$$-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} (r R_{k1}(r)) - \cancel{\frac{\hbar^2}{\mu r^2}} + \frac{\hbar^2}{\mu r^2} = E r R_{k1}(r)$$

$$\boxed{\begin{aligned} -\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} (r R_{k1}(r)) &= E r R_{k1}(r) \\ \checkmark -\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} u_{k1}(r) &= \frac{\hbar^2 k^2}{2\mu} u_{k1}(r) \end{aligned}}$$

$$u_{k1}''(r) = -k^2 u_{k1}(r)$$

$$\text{so } R_{k1}(r) = \alpha j_1(kr) + \beta n_1(kr) \quad (r > r_0)$$

$$0 = \alpha \bar{j}_1(kr_0) + \beta n_1(kr_0) \quad r=r_0$$

$$\Rightarrow \beta = \frac{\alpha}{\tan(kr_0)}$$

$$R_{k_1}(r) = \frac{\bar{j}_1(kr) - \tan(kr_0) n_1(kr)}{r}$$

In $r \rightarrow \infty$ limit, however,

$$\begin{aligned} R_{k_1}(r) &\sim \frac{\sin(kr - \frac{\pi}{2} - \delta_1)}{r} \\ &= \frac{\sin(kr) \cos(-\frac{\pi}{2} + \delta_1) + \cos(kr) \sin(-\frac{\pi}{2} + \delta_1)}{r} \end{aligned}$$

implying $\tan(kr_0) = -\tan(-\frac{\pi}{2} + \delta_1)$

$$\Rightarrow -kr_0 = -\frac{\pi}{2} + \delta_1$$

$$\text{or } \boxed{\delta_1 = \frac{\pi}{2} - kr_0}$$

$$V(r) = \alpha \delta(r-R)$$

We have that

$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + V(r) + \frac{\hbar^2 l(l+1)}{2\mu r^2} \right) u_{kl}(r) = \frac{\hbar^2 k^2}{2\mu} u_{kl}(r)$$

$l=0$ shift, $kr \ll 1$:

$$\left(\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + \alpha \delta(r-R) \right) u_{kl}(r) = \frac{\hbar^2}{2\mu} k^2 u_{kl}(r)$$

define $\alpha = -\frac{2m\alpha}{\hbar^2}$

$$\frac{d^2}{dr^2} u_{k0}(r) + \alpha \delta(r-R) u_{k0}(r) = -k^2 u_{k0}(r)$$

Solutions $r > R$, $r < R$ linear combinations of $j_0(kr)$, $n_0(kr)$

$\frac{d}{dr} u_{k0}(r)$ has discontinuity at $r=R$ (δ -function)

$$\left. \frac{d}{dr} u_{k0}(r) \right|_{r=R} = -\alpha u_{k0}(R) - k^2 \int_{R-\epsilon}^{R+\epsilon} dr u_{k0}(r)$$

$$u_{k_1}^{<R}(r) \sim \alpha \sin(kr) \quad r < R$$

$$u_{k_1}^{>R}(r) \sim \beta \sin(kr + \Delta) \quad r > R$$

At $r=R$, match continuity

$$\alpha \sin(kR) = \beta \sin(kR + \Delta) \quad (1)$$

and derivatives:

$$u_{k_1}^{<R}(r) \sim \alpha k \cos(kr)$$

$$u_{k_1}^{>R}(r) \sim \beta k \cos(kr + \Delta)$$

which both must be equal to $-\alpha u_{k_0}(R)$ at $r=R$:

$$\alpha k \cos(kR) = \beta k \cos(kR + \Delta) = -\alpha u_{k_0}(R)$$

Review of Rotations

Q1) Three-spin system (Quiz 2)

$$H = J(\vec{S}_1 \cdot \vec{S}_2 + \vec{S}_1 \cdot \vec{S}_3 + \vec{S}_2 \cdot \vec{S}_3) \quad s = \frac{1}{2} \text{ fermions}$$

Energy eigenvalues: observe that

$$(\vec{S}_1 + \vec{S}_2 + \vec{S}_3)^2 = 2(\vec{S}_1 \cdot \vec{S}_2 + \vec{S}_2 \cdot \vec{S}_3 + \vec{S}_1 \cdot \vec{S}_3) + S_1^2 + S_2^2 + S_3^2$$

$$\text{So } H = \frac{J}{2} (S_{\text{tot}}^2 - S_1^2 - S_2^2 - S_3^2)$$

$$= \frac{J}{2} (S_{\text{tot}}^2 - 3 \frac{3}{4} \hbar^2)$$

$$= \frac{J}{2} (S_{\text{tot}}^2 - \frac{9}{4} \hbar^2)$$

$$|\frac{1}{2} \pm \frac{1}{2}\rangle \quad \times 3$$

$$\frac{1}{2} \otimes 1 \rightarrow \frac{3}{2} \otimes \frac{1}{2}$$

$$\text{Now } S_{\text{tot}}^2 = \frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2}$$

$$= \frac{1}{2} \otimes (1 \otimes 0)$$

$$\frac{3}{2} \quad \frac{1}{2} \quad -\frac{1}{2} \quad -\frac{3}{2} \quad 4 \text{ states}$$

$$\frac{1}{2} - \frac{1}{2} \xleftarrow{\times 2} = \frac{1}{2}, \frac{3}{2} \xrightarrow{\quad}$$

$$S_{\text{tot}}^2 = \frac{1}{2} \Rightarrow \frac{3}{4} \hbar^2 - \frac{9}{4} \hbar^2 = -\frac{6}{4} \hbar^2 \quad (\times \frac{J}{2})$$

$$= -\frac{3}{4} \hbar^2 J \quad 2 \times 2\text{-fold} \quad (4 \text{ fold})$$

$$S_{\text{tot}}^2 = \frac{3}{2} \Rightarrow \frac{3}{2} \left(\frac{3}{2} + 1 \right) \hbar^2 - \frac{9}{4} \hbar^2 = \left(\frac{3}{4} \hbar^2 J \right) \quad 4\text{-fold} \quad (\text{for states})$$

both are 4-fold degenerate (4 states each)

c) 4-spin system:

$$H = \frac{J}{2} (S_{\text{tot}}^2 - S_1^2 - S_2^2 - S_3^2 - S_4^2)$$

$$\begin{aligned} 1001 &\rightarrow 2 \oplus 100 \quad \uparrow \text{3 states} \\ 100 &\rightarrow 1 \rightarrow \text{3 states} \\ 000 &\rightarrow 0 \quad \text{r2} \end{aligned}$$

$$\begin{aligned} \frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} &= (100) \oplus (100) \\ &= 2 \oplus 1 \oplus 0 \end{aligned}$$

$$2: 2 \ 1 \ 0 \ -1 \ -2$$

$$1: 1 \ 0 \ -1$$

Then $4 \times S_1^2 \rightarrow -4 \cdot \frac{3}{4} = -3\hbar^2$.

$$S_{\text{tot}}^2 = 2 \Rightarrow \hbar^2 2(2+1) = 6\hbar^2$$

$$= 1 \Rightarrow \hbar^2 1(1+1) = 2\hbar^2$$

$$= 0 \Rightarrow 0$$

Thus $E_2 = \frac{3}{2} J \hbar^2 \leftarrow 5\text{-fold}$

$$E_1 = -\frac{1}{2} J \hbar^2 \leftarrow 3 \cdot 3 = 7\text{-fold}$$

$$E_0 = -\frac{3}{2} J \hbar^2 \leftarrow 2\text{-fold}$$

with degeneracy

The idea of degeneracy lies in the multiple addition of angular momentum states:

$$\begin{aligned}
 2. \quad \frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} &= \frac{1}{2} \otimes (1 \otimes 0) \\
 &= \frac{1}{2} \otimes 1 + \frac{1}{2} \otimes 0 \\
 &= \frac{3}{2} \oplus \frac{1}{2} \oplus \frac{1}{2} \\
 &\quad \swarrow \quad \searrow \downarrow \\
 &\quad \frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2} \quad \frac{1}{2}, -\frac{1}{2}
 \end{aligned}$$

Must account for all addition sums:

$$\begin{aligned}
 \frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} &\rightarrow (1 \otimes 0) \otimes (1 \otimes 0) \\
 &\rightarrow 1 \otimes 1 + 2 \cdot 1 \otimes 0 + 0 \otimes 0 \\
 &\rightarrow 2 \otimes 1 \otimes 0 + 2 \cdot 1 + 0
 \end{aligned}$$

$$\begin{array}{lcl}
 1\text{-fold} \rightarrow & 2 & 1 \quad 0 \quad -1 \quad 2 \quad 2x \\
 3 \times (1+2) = 9\text{-fold} \rightarrow & 1 & 0 \quad -1 \quad 1 \quad 0 \quad -1 \\
 & 0 & \leftarrow \text{2 fold} \rightarrow 0
 \end{array}$$

Rotation of a Spin State (Hw)

$|+-\rangle$ state:

$$S_{1z} |+-\rangle = \frac{\hbar}{2} |+-\rangle$$

$$S_{2z} |+-\rangle = -\frac{\hbar}{2} |+-\rangle$$

Rotate 90° about $\hat{x}$ axis:

We have separate rotations about $\hat{x}$ axis:

$$\begin{aligned} U\left(\frac{\pi}{2}, \hat{x}\right) &= \cos \frac{\pi}{4} \sigma_0 + i \sin \frac{\pi}{4} \sigma_x \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \end{aligned}$$

$$\text{on } |+\rangle_z \rightarrow U \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} = \frac{1}{\sqrt{2}} (|+\rangle + i|-\rangle)$$

$$\text{on } |-\rangle_z \rightarrow U \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} (i|+\rangle + |-\rangle)$$

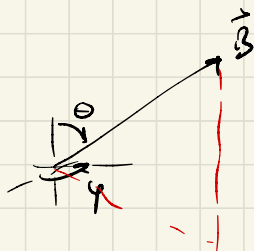
$$\text{Hence } |+\rangle \otimes |-\rangle \rightarrow U|+\rangle \otimes U|-\rangle$$

$$\rightarrow \frac{1}{2} (|+\rangle + i|-\rangle) \otimes (i|+\rangle + |-\rangle)$$

$$= \frac{1}{2} (i|++\rangle + |+-\rangle - |-\rangle + i|--\rangle)$$

Spin Precession

$$\vec{B} \rightarrow \vec{B}(\theta, \varphi)$$



$$\vec{B} = (B \sin \theta \cos \varphi, B \sin \theta \sin \varphi, B \cos \theta)$$

At $t=0$ electron has spin $\parallel$ to $\vec{B}$.

$$|\psi\rangle = A|+\rangle + B|-\rangle.$$

Find coefficients A, B in $\hat{S}_z$ -basis.

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = H |\psi\rangle$$

$$= -\frac{q}{m} \vec{B} \cdot \vec{S} |\psi\rangle$$

$$\omega = \frac{qB}{2m}$$

$$e^{i\omega t \hat{n} \cdot \vec{\sigma}}$$

$$\text{Then } |\psi(t)\rangle = e^{i\frac{q}{2m} \vec{B} \cdot \vec{\sigma} t} |\psi(0)\rangle$$

$$\vec{B} = B \hat{n} \cdot \vec{\sigma}$$

$$\text{Now } \vec{B} \cdot \vec{\sigma} = B \left(\begin{pmatrix} 0 & \sin \theta \cos \varphi \\ \sin \theta \cos \varphi & 0 \end{pmatrix} + \begin{pmatrix} 0 & -i \sin \theta \sin \varphi \\ i \sin \theta \sin \varphi & 0 \end{pmatrix} + \begin{pmatrix} \cos \theta & 0 \\ 0 & -\cos \theta \end{pmatrix} \right)$$

$$= B \begin{pmatrix} \cos \theta & \sin \theta e^{-i\varphi} \\ \sin \theta e^{i\varphi} & -\cos \theta \end{pmatrix} \quad \text{let } \omega = \frac{qB}{2m}$$

$$\text{Then } |\psi(t)\rangle = |\psi(0)\rangle \left[\cos \omega t \sigma_0 + i \sin \omega t \begin{pmatrix} \cos \theta & \sin \theta e^{i\varphi} \\ \sin \theta e^{-i\varphi} & -\cos \theta \end{pmatrix} \right]$$

What's $|\psi(0)\rangle$ though? Should be $\parallel$ to $\vec{B}$: spin (up?)

$$|\psi(0)\rangle = \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\varphi/2} \\ \sin \frac{\theta}{2} e^{i\varphi/2} \end{pmatrix}$$

$$|\psi(t)\rangle = \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\varphi/2} \\ \sin \frac{\theta}{2} e^{i\varphi/2} \end{pmatrix} \begin{pmatrix} \cos \omega t + i \sin \omega t \cos \theta & i \sin \omega t \sin \theta e^{i\varphi} \\ i \sin \omega t \sin \theta e^{-i\varphi} & \cos \omega t - i \sin \omega t \cos \theta \end{pmatrix}$$

$$= \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\varphi/2} \cos \omega t - i \cos \frac{\theta}{2} e^{-i\varphi/2} \sin \omega t \cos \theta + \cos \frac{\theta}{2} e^{-i\varphi/2} i \sin \omega t \sin \theta e^{i\varphi} \\ i \sin \frac{\theta}{2} e^{i\varphi/2} \sin \omega t \sin \theta e^{-i\varphi} + \sin \frac{\theta}{2} e^{i\varphi/2} \cos \omega t - i \sin \frac{\theta}{2} e^{i\varphi/2} \sin \omega t \cos \theta \end{pmatrix}$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$= \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\varphi/2} \cos \omega t + i \cos \frac{3\theta}{2} e^{-i\varphi/2} \sin \omega t - i \cos \frac{\theta}{2} \sin^2 \frac{\theta}{2} \sin \omega t e^{-i\varphi/2} \\ + 2i \cos \frac{2\theta}{2} e^{i\varphi/2} \sin \omega t \sin \frac{\theta}{2} \\ 2i \sin^2 \frac{\theta}{2} e^{-i\varphi/2} \sin \omega t + \sin \frac{\theta}{2} e^{i\varphi/2} \cos \omega t - i \sin \frac{\theta}{2} \cos^2 \frac{\theta}{2} e^{i\varphi/2} \sin \omega t \\ + i \sin^3 \frac{\theta}{2} e^{i\varphi/2} \sin \omega t \end{pmatrix}$$

Not sure what to do w/ this? If $t=0$,

$$= \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\varphi/2} \\ \sin \frac{\theta}{2} e^{i\varphi/2} \end{pmatrix}$$

then

$$A = \cos \frac{\theta}{2} e^{-i\varphi/2}$$

$$B = \sin \frac{\theta}{2} e^{i\varphi/2}$$

Back to Perturbation Theory

Q2) Particle in a box $[-\frac{L}{2}, \frac{L}{2}]$.

Subject to a potential $W(x) = \alpha x^2$.

For $0 < \alpha \ll 1$, estimate energy eigenvalues.

Hamiltonian $H = \frac{p_x^2}{2m} + V(x) + W(x)$

$$V(x) = \begin{cases} 0 & x \in [-\frac{L}{2}, \frac{L}{2}] \\ \infty & \text{otherwise} \end{cases}$$

Stationary solutions $e^{-i \frac{E_0 t}{\hbar}}$

$$E \psi(x) = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right) \psi(x)$$

$$V(x) = 0 \quad \text{implies} \quad \psi''(x) = -\frac{2mE}{\hbar^2} \psi \quad \omega = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\text{or } \psi(x) = A \cos \omega x + B \sin \omega x$$

$$A + \psi(\pm \frac{L}{2}) = 0: \quad \text{odd condition}$$

$$\frac{\omega L}{2} = n\pi$$

$$0 = A \cos \omega \frac{L}{2} + B \sin \omega \frac{L}{2}$$

$$\frac{2mEL^2}{\hbar^2 4} = n^2 \pi^2$$

$$0 = A \cos \omega \frac{L}{2} - B \sin \omega \frac{L}{2}$$

$$E = \frac{2n^2 \pi^2 \hbar^2}{mL^2}$$

$$A \cos \omega \frac{L}{2} = B \sin \omega \frac{L}{2}$$

$$A = -B \frac{\sin \omega \frac{L}{2}}{\cos \omega \frac{L}{2}}$$

$$0 = -B \sin \frac{\omega L}{2} - B \sin \frac{\omega L}{2}$$

$$\Rightarrow \frac{\omega L}{2} = n\pi$$

$$\text{or } \frac{2mE}{\hbar^2} \frac{1}{4} L^2 = n^2 \pi^2$$

$$\omega = \sqrt{\frac{2m \cdot 2n^2 \pi^2 \hbar^2}{4L^2}}$$

$$\omega = \frac{2n\pi}{L}$$

$$\text{or } E_n = \frac{2n^2 \pi^2 \hbar^2}{mL^2}$$

energy eigenvalues.

What about eigenstates?

$$\psi(x) = A \cos\left(\frac{2n\pi x}{L}\right) + B \sin\left(\frac{2n\pi x}{L}\right)$$

$$\psi\left(-\frac{L}{2}\right) = 0 = A \cos(4n\pi) + B \sin(4n\pi)$$

but $\sin(4n\pi) = 0$ so then $A = 0$.

$$\psi(x) = A_n \sin\left(\frac{2n\pi x}{L}\right)$$

and normalized $\int_{-\frac{L}{2}}^{\frac{L}{2}} A_n^2 \sin^2\left(\frac{2n\pi x}{L}\right) dx = 1$

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x)$$

$$\frac{1}{2} \int_{-\frac{L}{2}}^{\frac{L}{2}} A_n^2 dx - \frac{1}{2} \int_{-\frac{L}{2}}^{\frac{L}{2}} \cos\left(\frac{4n\pi x}{L}\right) dx = 1$$

$$- \frac{L}{4n\pi} \sin\left(\frac{4n\pi x}{L}\right) \Big|_{-\frac{L}{2}}^{\frac{L}{2}} = 0$$

$$\text{So } \frac{1}{2} A_n^2 \cdot \left(\frac{L}{2} - \left(-\frac{L}{2}\right) \right) = 1$$

$$\frac{L}{2} A_n^2 = 1$$

$$A_n = \sqrt{\frac{2}{L}}$$

So

$$\psi_n^0(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{2n\pi x}{L}\right)$$

Ground state,
unperturbed
energies.

We now consider the perturbation term: αx^2 .

No E 's are degenerate so

$$E_n(x) = E_n^0 + \alpha \langle \psi_n^0 | x^2 | \psi_n^0 \rangle + \alpha^2 \sum_{n \neq m} \frac{|\langle \psi_n^0 | x^2 | \psi_m^0 \rangle|^2}{E_n^0 - E_m^0}$$

Term by term: $\alpha \langle \psi_n^0 | x^2 | \psi_n^0 \rangle$

$$= \alpha \int_{-\frac{L}{2}}^{\frac{L}{2}} dx \frac{2}{L} \sin^2\left(\frac{2n\pi x}{L}\right) x^2$$

$$= \frac{2\alpha}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} dx \frac{1}{2} (1 - \cos(\frac{4n\pi x}{L})) x^2$$

$$= \frac{\alpha}{L} \left[\int_{-\frac{L}{2}}^{\frac{L}{2}} dx x^2 - \int_{-\frac{L}{2}}^{\frac{L}{2}} dx \cos\left(\frac{4n\pi x}{L}\right) x^2 \right]$$

$$\frac{1}{3} \left(\frac{L^3}{8} - \left(-\frac{L^3}{8}\right) \right) = \frac{1}{12} L^3$$

$$\int_{-\frac{L}{2}}^{\frac{L}{2}} \cos \omega x \, x^2 dx \rightarrow \begin{array}{rcl} & 0 & \frac{L}{2} \\ + & x^2 & \cos \omega x \\ - & 2x & \frac{1}{\omega} (-\sin \omega x) \Big|_{-\frac{L}{2}}^{\frac{L}{2}} \\ + & 2 & \frac{1}{\omega^2} \cos \omega x \Big|_{-\frac{L}{2}}^{\frac{L}{2}} \\ - & 0 & \frac{1}{\omega^3} \sin \omega x \Big|_{-\frac{L}{2}}^{\frac{L}{2}} \end{array}$$

any $\sin(2\omega) \Big|_{-\frac{L}{2}}^{\frac{L}{2}} \rightarrow 0$
by boundary conditions

$$-\frac{x^2}{\omega} \sin(\omega x) \Big|_{-\frac{L}{2}}^{\frac{L}{2}} = -\frac{1}{\omega} \left(\frac{L^2}{4} \sin\left(\frac{4n\pi L}{2L}\right) + \frac{L^2}{4} \sin\left(\frac{4n\pi L}{2L}\right) \right)$$

$$= -\frac{L^2}{2\omega} \sin(2n\pi) \rightarrow 0$$

$$\frac{2x}{\omega^2} \cos \omega x \Big|_{-\frac{L}{2}}^{\frac{L}{2}} = \frac{L}{\omega^2} \cos\left(\frac{4n\pi L}{2L}\right) - \left(-\frac{L}{\omega^2}\right) \cos\left(\frac{4n\pi L}{2L}\right)$$

$$= \frac{2L}{\omega^2} \cos(4n\pi) \rightarrow 2$$

$$= \frac{2L}{\frac{4^2 n^2 \pi^2}{L^2}} = \frac{L^3}{8n^2 \pi^2}$$

$$= \frac{\alpha}{L} \left[\frac{1}{12} L^3 - \frac{L^3}{8n^2 \pi^2} \right]$$

$$E_n^{(1)} = \frac{\alpha}{4} L^2 \left(\frac{1}{3} - \frac{1}{2n^2 \pi^2} \right)$$

first order
correction.

2nd order :

$$\alpha^2 \sum_{n \neq m} \frac{|\langle \psi_n^0 | x^2 | \psi_m^0 \rangle|^2}{E_n^0 - E_m^0}$$

Note $E_n^0 = \frac{2n^2 \pi^2 \hbar^2}{mL^2}$

$$\text{so } E_n^0 - E_m^0 = \frac{2\pi^2 \hbar^2}{mL^2} (n^2 - m^2)$$

$$|\langle \psi_n^0 | x^2 | \psi_m^0 \rangle|^2 = \left| \int_{-\frac{L}{2}}^{\frac{L}{2}} \underbrace{\frac{2}{L} \cdot \sin\left(\frac{2n\pi x}{L}\right) \sin\left(\frac{2m\pi x}{L}\right)}_{= I} x^2 dx \right|^2$$

Do I : Sineoid orthogonality :

$$\frac{2}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} \sin\left(\frac{2n\pi x}{L}\right) \sin\left(\frac{2m\pi x}{L}\right) dx = \begin{cases} 1 & n=m \neq 0 \\ 0 & n \neq m \end{cases}$$

So the whole 2nd order correction is 0.

$$E_n(\alpha) = \frac{2n^2 \pi^2 \hbar^2}{mL^2} + \frac{\alpha L^2}{4} \left(\frac{1}{3} - \frac{1}{2n^2 \pi^2} \right) + \alpha^2 \cdot 0 + o(\alpha^3)$$

Not correct but very close, I don't think you solved the ODE properly

Q3) PT HW
Electron in Hydrogen in nucleus $\vec{E}$ -field.

Spin-orbit coupling : relativity $\Rightarrow e^-$ feels $\vec{B}$ and $\vec{E}$ fields.

$$H_{so} = \frac{e^2}{2m^2 c^2 r^3} \vec{S} \cdot \vec{L}$$

Take $H = H_0 + H_{so}$

$$= \frac{\vec{p}^2}{2\mu} + \frac{e^2}{|r|} + H_{so} \dots$$

2) Neglect H_{so} at first - states w/ $n=2$, find degeneracy.

$$H = -\frac{\hbar^2}{2\mu} \nabla^2 - \frac{e^2}{|r|}$$

Stationary states $H \psi_{n\ell m} = E_{n\ell} \psi_{n\ell m}$

$$\psi_{n\ell m} = R_{n\ell}(r) Y_{\ell}^m(\theta, \phi)$$

Radial eqn gives

$$\left[\underbrace{-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + \frac{\hbar^2 \ell(\ell+1)}{2\mu r^2}}_{V_{\ell, \text{eff}}(r)} - \frac{e^2}{r} \right] u_{n\ell}(r) = E_{n\ell} u_{n\ell}(r)$$

We find $E_{n\ell} = \frac{-E_1}{(n_r + \ell)^2} \rightarrow E_n = -\frac{E_1}{n^2}$ (in principle quantum #)

For $n=2$, $E_2 = -\frac{E_1}{4}$ so we wish to find the degeneracy of this state. $E_1 = \frac{1}{2} \alpha^2 \mu c^2$ ($\alpha = \frac{1}{137}$ fine structure)

$$E_2 = -\frac{e^2}{8a_0} \quad (2a_0 : \text{Bohr Radius})$$

We then need to find the number of states st.

$$n_r + l = 2 : \quad (n_r \text{ being orbital \#})$$

$$(n_r \neq 0)$$

$$\left. \begin{array}{ll} n_r=1 & l=1 \\ n_r=2 & l=0 \end{array} \right\} E_2 = -\frac{e^2}{8a_0}$$

$$\text{So } l=1, 0. \quad \text{Since } \vec{S} \cdot \vec{L} = \frac{1}{2} (\vec{J}_{tot}^2 - \vec{S}^2 - \vec{L}^2),$$

$$\text{we have } \frac{1}{2} \otimes (1 \otimes 0) = \frac{3}{2}, \frac{1}{2}, \frac{1}{2}$$

$$\left. \begin{array}{l} \frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}, \\ \frac{1}{2}, -\frac{1}{2} \\ \frac{1}{2}, -\frac{1}{2} \end{array} \right\} 4 + 4 = 8$$

States
degenerate

for a spin- $\frac{1}{2}$
particle.

Now treat H_{so} as a small perturbation.

$$H_{so} = \frac{e^2}{2m^2c^2r^3} \vec{S} \cdot \vec{L}$$

$$\frac{1}{2} \otimes 1 = \frac{1}{2} \oplus \frac{3}{2}$$

$$= \frac{e^2}{4m^2c^2r^3} (\vec{J}^2 - \vec{S}^2 - \vec{L}^2)$$

We have the angular momentum states $|l, m_l; \frac{1}{2}, m_s\rangle$

$$l=0 \quad \frac{3}{4}\hbar^2, \quad \frac{15}{4}\hbar^2 \quad l=1$$

Then $\frac{e^2}{4m^2c^2r^3} \left[\langle l, m_l; \frac{1}{2}, m_s | \vec{J}^2 | l, m_l; \frac{1}{2}, m_s \rangle \right.$

$$- \langle l, m_l; \frac{1}{2}, m_s | \vec{S}^2 | l, m_l; \frac{1}{2}, m_s \rangle$$

$$\frac{3}{4}\hbar^2$$

$$- \langle l, m_l; \frac{1}{2}, m_s | \vec{L}^2 | l, m_l; \frac{1}{2}, m_s \rangle \Big]$$

4 states of degeneracy like here.

0 if $l=0$,
 $2\hbar^2$ if $l=1$ (4 states of degeneracy like here)

Construct a matrix and diagonalize.

Consider 3 pairs (j, L, S) together (S is the same $= \frac{1}{2}$)

$$j = \frac{3}{2}, \frac{1}{2} \quad (\text{total } j) \rightarrow 4 \text{ states}$$

$$l = 1, 0 \rightarrow 4 \text{ states}$$

$$s = \frac{1}{2}$$

$$L^2 - 3 = \frac{12}{4} \hbar^2 = 3 \hbar^2$$

we have $j = \frac{3}{2}, J^2 = \frac{15}{4} \hbar^2$

$$l = 1, -L^2 = -2 \hbar^2$$

$$s = \frac{1}{2}, -S^2 = \frac{3}{4} \hbar^2 \quad \text{always}$$

$$+ \frac{\hbar^2}{4}$$

$$j = \frac{1}{2}, J^2 = \frac{3}{4} \hbar^2$$

$$l = 1, -L^2 = -2 \hbar^2$$

$$+ \frac{-2 \hbar^2}{-2 \hbar^2}$$

$$l = 0, 0$$

$$+ \frac{0}{0}$$

Hence

$$\begin{pmatrix} \hbar^2 & & \\ & -2 \hbar^2 & \\ & & 0 \end{pmatrix}$$

and these are the energy corrections.

2014 Exam

Hydrogen atom $|n\ell m \pm\rangle$

b) Spinor representation of $e^{-i\frac{\mu}{2}\sigma_y} |211+\rangle$

$$= \left(\cos \frac{\mu}{2} \sigma_0 - i \sin \frac{\mu}{2} \sigma_y \right) |211+\rangle$$

$$= \begin{pmatrix} \cos \frac{\mu}{2} & -\sin \frac{\mu}{2} \\ \sin \frac{\mu}{2} & \cos \frac{\mu}{2} \end{pmatrix} \begin{pmatrix} \psi_{211} \\ 0 \end{pmatrix}$$

$$= |211+\rangle \begin{pmatrix} \cos \frac{\mu}{2} \\ \sin \frac{\mu}{2} \end{pmatrix}$$

$$= |211+\rangle \otimes \left[|+\rangle \cos \frac{\mu}{2} + |-\rangle \sin \frac{\mu}{2} \right]$$

Q4) Distinguishable 2-spin system

$$H = \frac{1}{2} A (S_{\text{tot}}^2 - S_1^2 - S_2^2)$$

$$\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$$

that is,

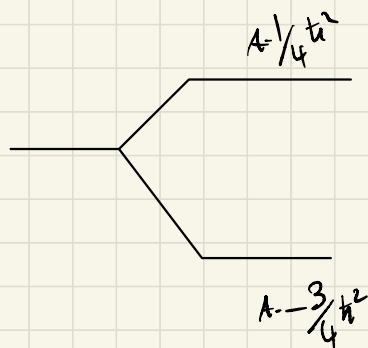
$$\begin{array}{cc} 1 & 0 & -1 & \text{States} \\ & 0 & & \text{States} \end{array}$$

$$S_{1/2}^2 |s_1, m_{s_1}\rangle = \frac{3\hbar^2}{4}$$

$$S_{\text{tot}}^2 = 2\hbar^2 \text{ or } 0$$

3 fold degenerate

nondegenerate



Eigenstates w/ $-A \frac{3}{4} \hbar^2$ and $A \frac{1}{4} \hbar^2$ energies.

nondegenerate

3 fold degenerate

b) Physical interpretation:

Negative energy corresponds to Spin-alignment

positive energy corresponds to Spin-anti alignment

Argument for $\frac{d\sigma}{d\Omega}$. We have $\psi = e^{ikz} + f_k(\theta) \frac{e^{ikr}}{r}$.

$$\text{Now } \frac{\# \text{ particles}}{\text{unit time}} = dN$$

$$\text{But } \text{flux } (I) = \frac{\# \text{ particles}}{\text{unit time} \times \text{unit area}}$$

$$\text{So } I = A dN = r^2 \vec{j} d\Omega$$

$$\text{if this area is } A = r^2 d\Omega \text{ (solid angle)}$$

$$\text{then } I = r^2 \vec{j} d\Omega.$$

$$\text{The if } \vec{j} = \frac{\hbar}{m} \frac{|f_k(\theta)|^2}{r^2} \nabla(kr)$$

$$= \frac{\hbar}{m} \frac{|f_k(\theta)|^2}{r^2} \hat{r} k$$

$$\text{implying that } I_{\text{at}} = \frac{\hbar}{m} |f_k(\theta)|^2 k d\Omega$$

$$I_{\text{incident}} = d\sigma \cdot K \frac{\hbar^2 k}{m}$$

$$\text{or } \frac{\hbar^2}{m} |f_k(\theta)|^2 k d\Omega = K \frac{\hbar^2 k}{m} d\sigma$$

$$\text{or } \frac{d\epsilon}{ds} = \|f_K(\theta, \varphi)\|^2$$